

References in the Koine Greek Bible



Presented by Wes Stevick & Isaac Anderson

Our mission.

Discover how New Testament Authors use
the Old Testament.

Our tools.

Word2Vec

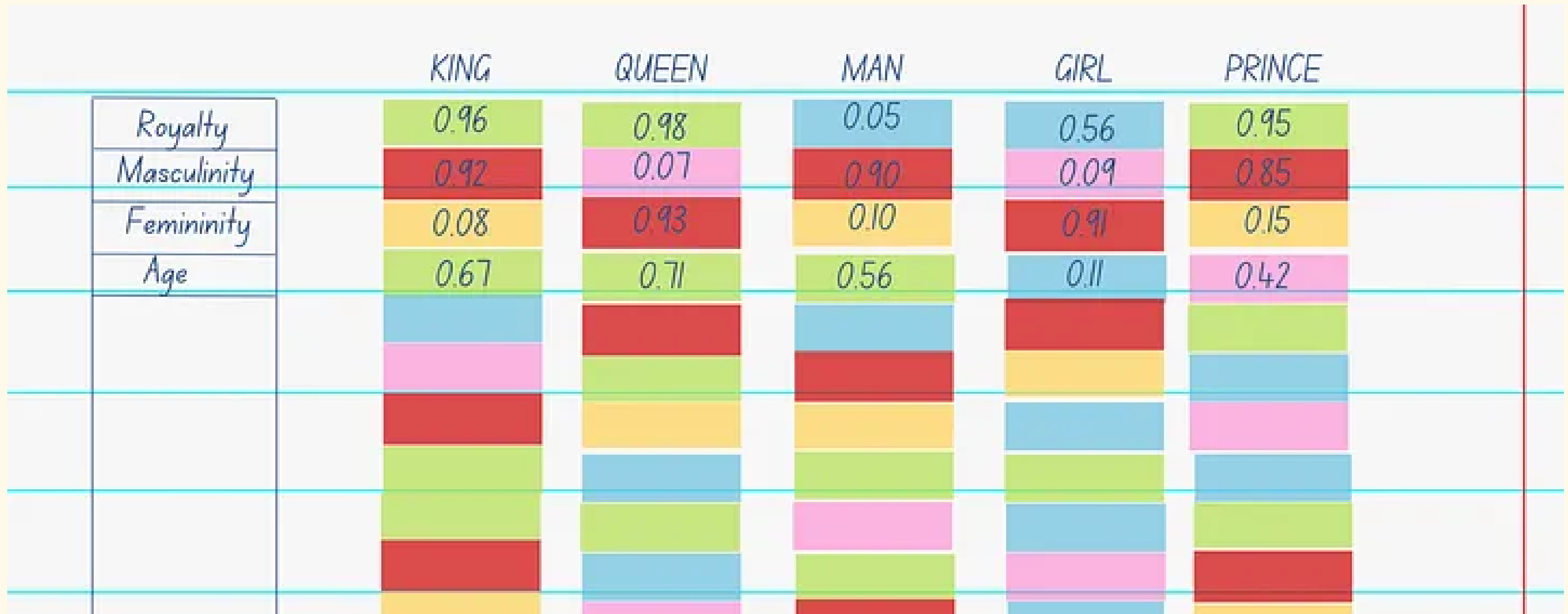
Sliding Window Search

→ Word2Vec

Sliding Window Search

What is Word2Vec?

- Turning words into vectors
- Visualising the different attributes of the word



Why Word2Vec Didn't work

66 books

30,782 verses

612,952 words

52,740 unique words

The top 10 words make up 39%
and are all semantically
meaningless with the exception
of LORD.

Why Word2Vec Didn't work

Loss per word:

Unstable: **20-40**

Learning structure: **5-10**

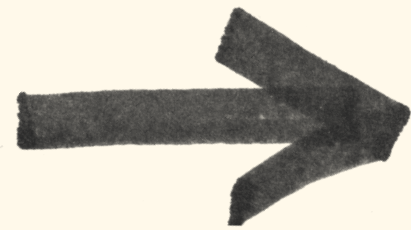
Converged: **2-6**

Our loss after 100 epochs: 61,292,500

$61,292,500 / 46,067 = \mathbf{1,330}$ per word

$$61,292,500 / 46,067 = 1,330 \text{ per word}$$

Semantically
Meaningless



Unstable

Word2Vec

→ Sliding Window Search

Our Other Approach: Sliding Window

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“Lorem ipsum dolor sit” != “Ut enim Lorem ipsum”

Our Other Approach: Sliding Window

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“Lorem ipsum dolor sit” != “enim Lorem ipsum dolor”

Our Other Approach: Sliding Window

4
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"Lorem ipsum dolor sit" != "Lorem ipsum dolor sit"



OT NT
(0, 4, 2, 4)

Our Other Approach: Sliding Window

References Database

(481601, 178, 5),

(481602, 179, 5),

(479354, 574, 5),

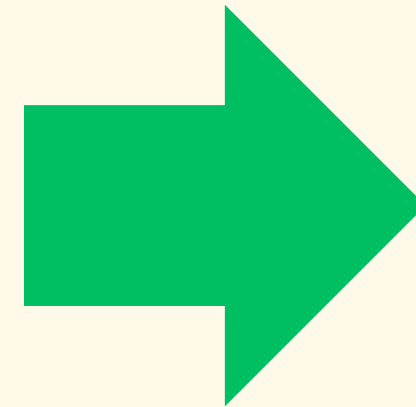
(482957, 575, 5),

(486353, 607, 5),

(479354, 633, 5),

(482957, 634, 5),

(482957, 695, 5),



(41601, 178, 6)

The Results

The Results

The Most quoted verses in the Bible

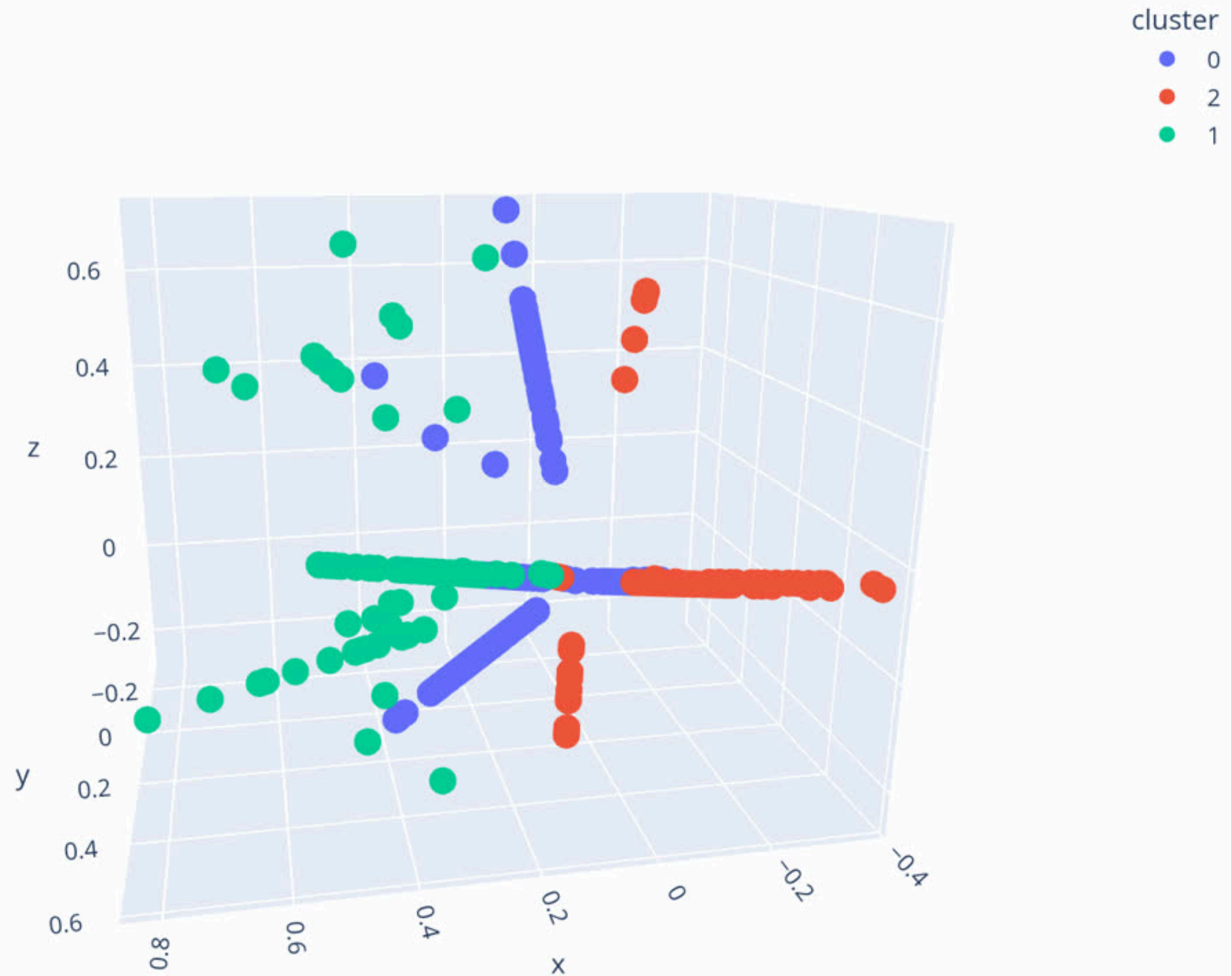
And I will establish his seed forever, and his throne as the days of heaven.	Psalms 89:29	22 times
He has scattered abroad, he has given to the poor; his righteousness endures forever; his horn shall be exalted in glory.	Psalms 112:9	22 times

The Results

**Words that occur most in references
(compared to all OT)**

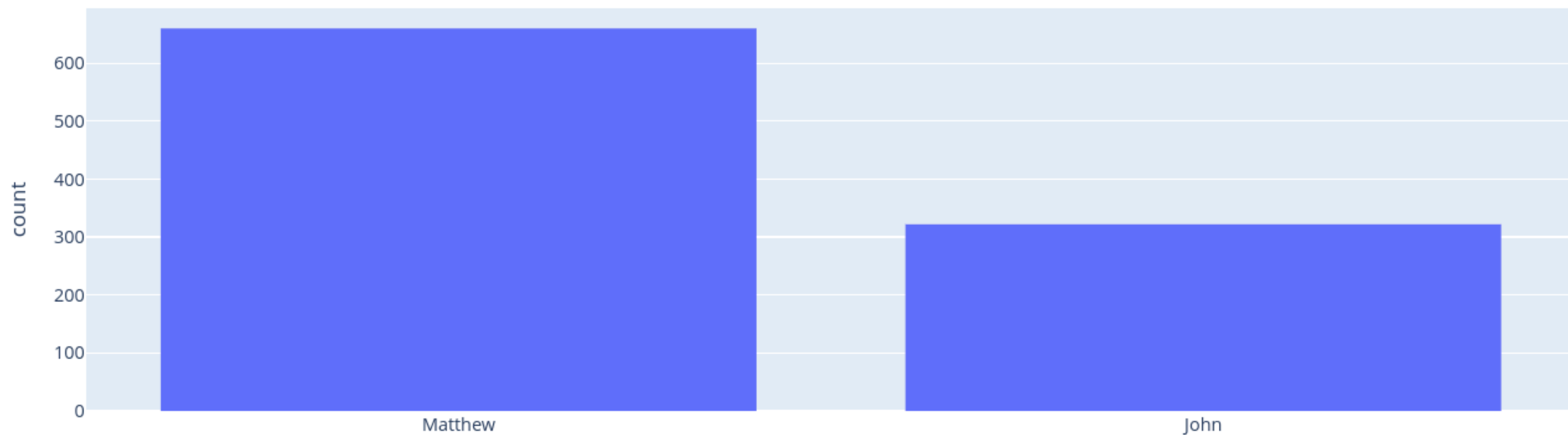
- 1. “That” (loosely)**
- 2. “Father”**
- 3. “God”**
- 4. “You”**
- 5. “In”**

The Results



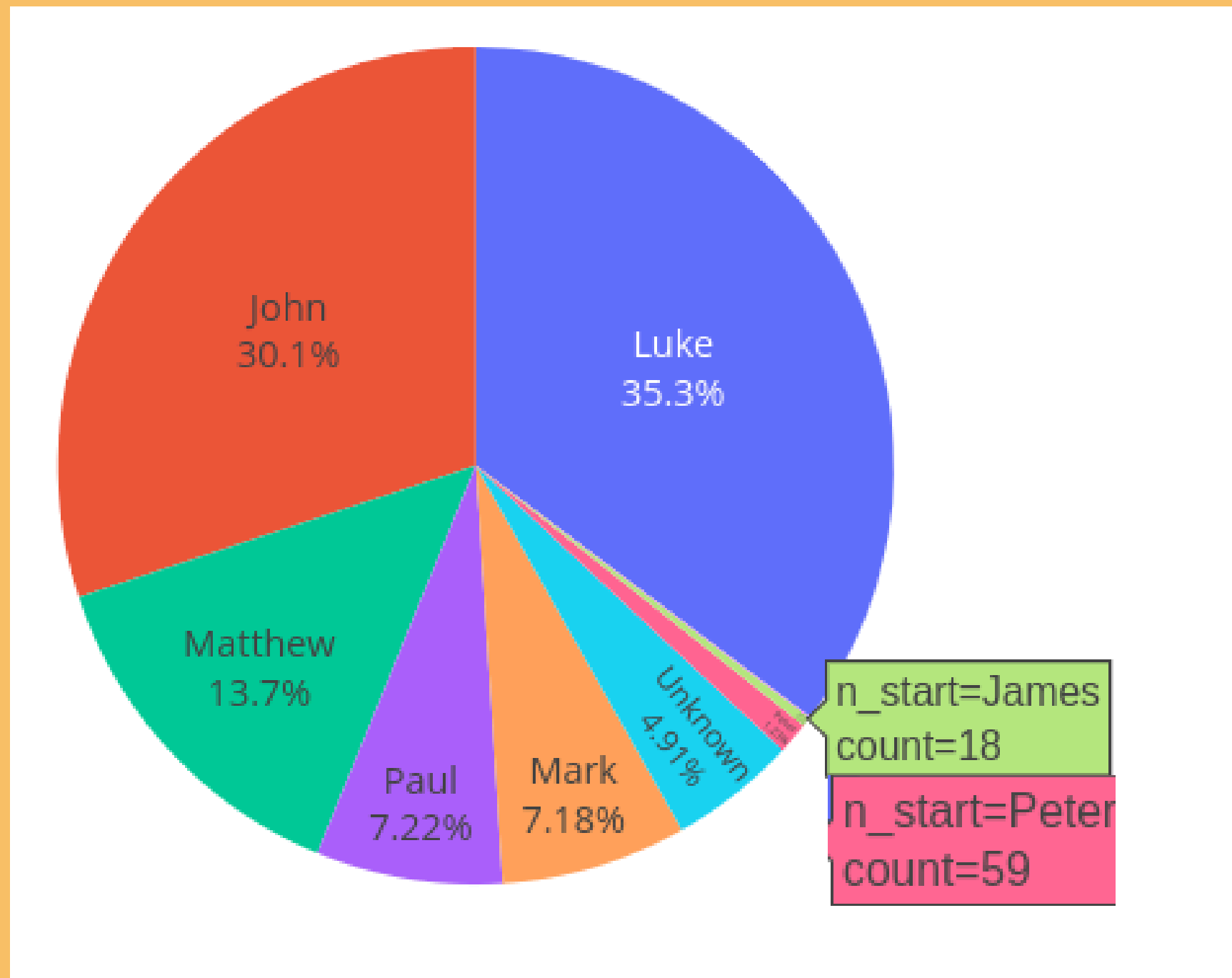
The Results

Which has more Allusions, Matthew or John?



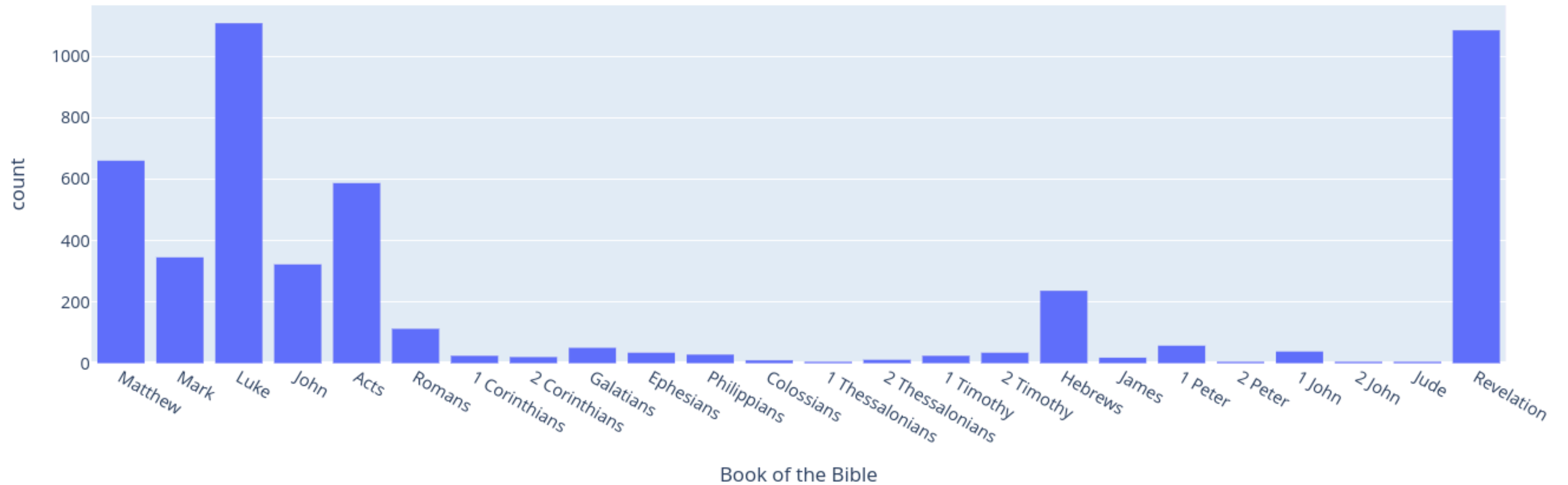
The Results

Allusions by each Author (%)



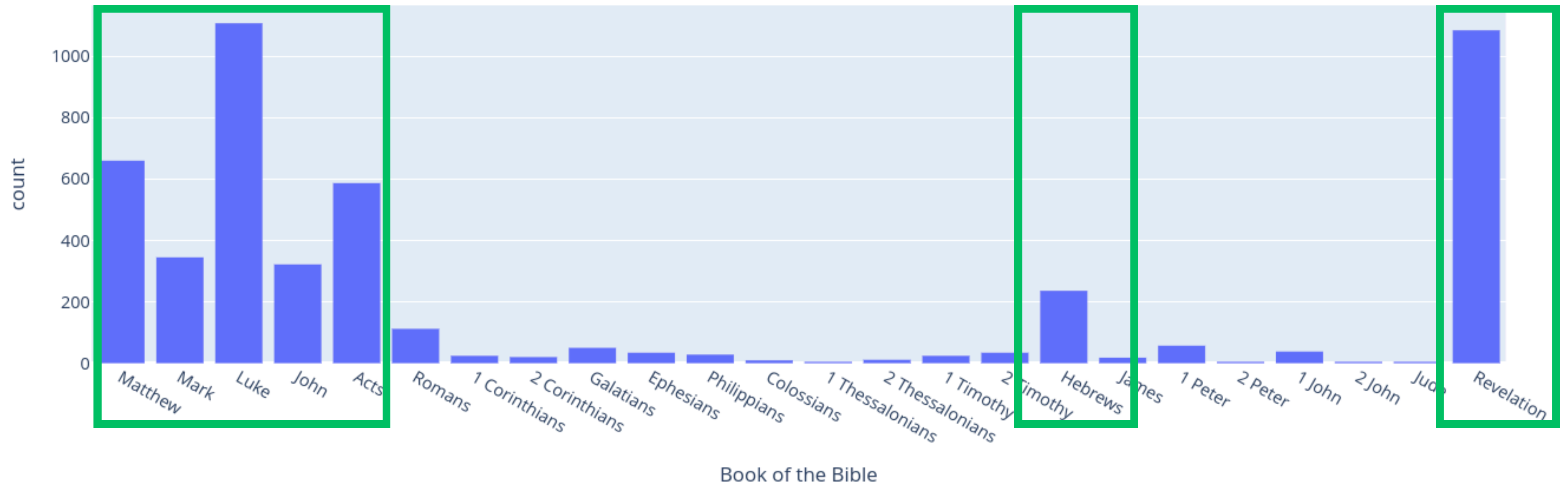
The Results

Number of References in each New Testament Book



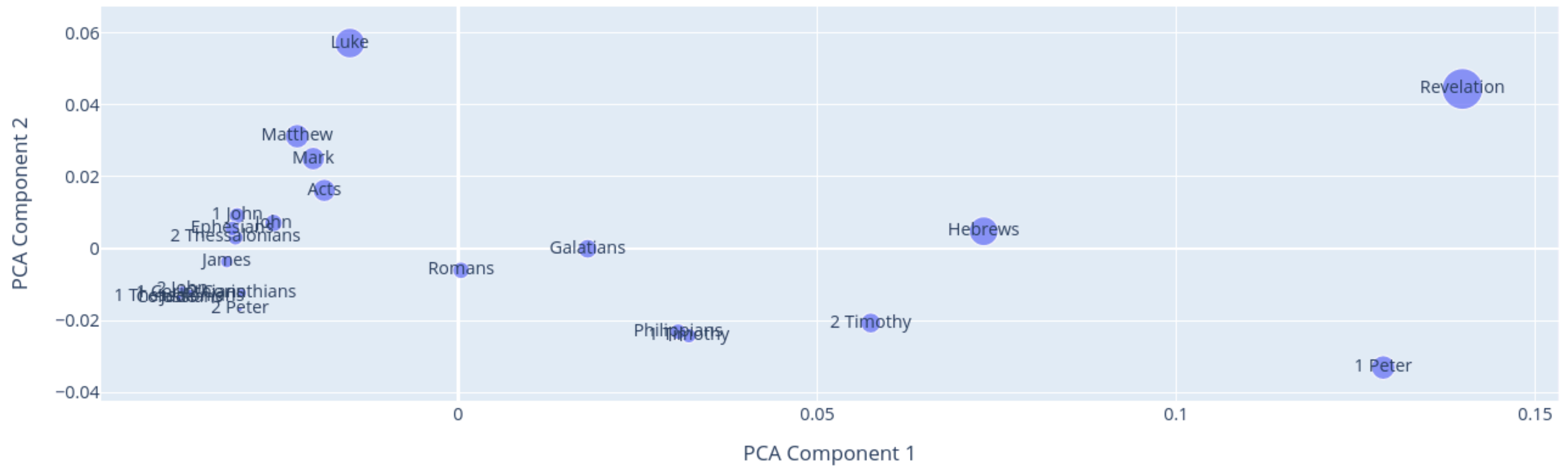
The Results

Number of References in each New Testament Book



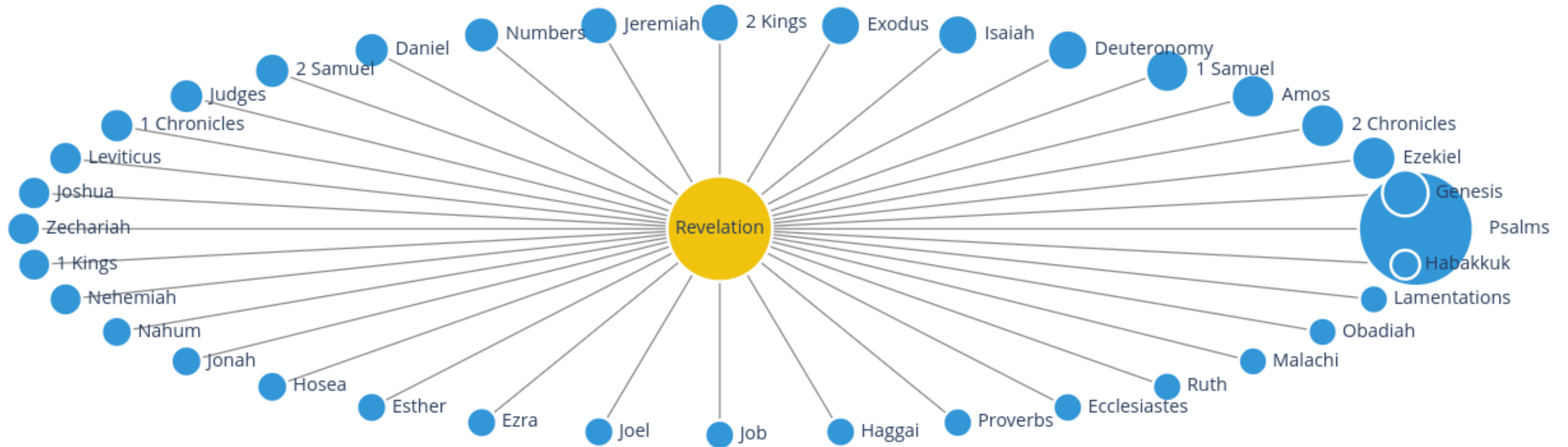
The Results

New Testament Use of Old Testament books

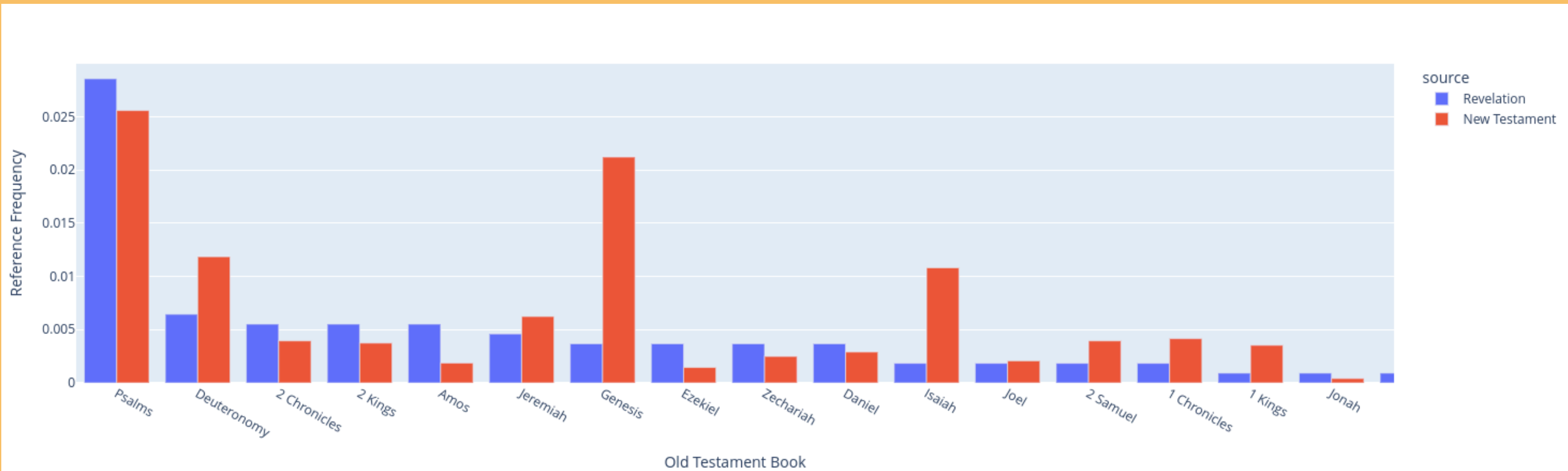


The Results

All the References in Revelations



The Results



THE END.